

A 13-VERTEX GRAPH WITH NO 4-CYCLE WHOSE WORST ORIENTATION BEATS THE MATCHING BOUND

ABSTRACT. For a graph G let $\text{mold}(G)$ be the largest value of the locating-domination number $\gamma_{\text{LD}}(D)$ over all orientations D of G . Bousquet, Deschamps, Lehtilä and Parreau proved that $\text{mold}(G) \leq n - \alpha'(G)$ whenever G has no C_4 subgraph, where $n = |V(G)|$ and α' is the matching number, and asked whether the inequality is ever strict (Open problem 37 in the e-print of their paper). It is. We exhibit a graph G on 13 vertices and 22 edges with no 4-cycle, graph6 string `L??E@agXAp@wdC`, for which

$$\text{mold}(G) = 6 < 7 = n - \alpha'(G).$$

The lower bound $\text{mold}(G) \geq 6$ is exhibited by one orientation and is checkable by hand; the upper bound $\text{mold}(G) \leq 6$ is a finite statement about all $2^{22} = 4\,194\,304$ orientations. We do not claim that G is smallest or unique.

1. THE PROBLEM

All graphs are finite, simple and undirected; an *orientation* D of G assigns to each edge one of its two directions. For $u \in V(D)$ write $N_D^-(u)$ for the set of in-neighbours of u . Following Bousquet, Deschamps, Lehtilä and Parreau [1], a set $S \subseteq V(D)$ is *locating-dominating* in the oriented graph D when the sets

$$I_D(u) = N_D^-(u) \cap S \quad (u \in V(D) \setminus S)$$

are all non-empty and pairwise distinct; $\gamma_{\text{LD}}(D)$ denotes the least size of such a set, which exists because $S = V(D)$ is vacuously locating-dominating. The parameter studied in [1, Section 4] is the *worst* orientation value

$$\text{mold}(G) = \max\{\gamma_{\text{LD}}(D) : D \text{ an orientation of } G\}.$$

Say that G is *without* C_4 if G has no 4-cycle as a subgraph, induced or not; equivalently, no two vertices of G – adjacent or not – have two common neighbours. This is the hypothesis of [1, Lemma 34], which states that

$$(1) \quad \text{mold}(G) \leq n - \alpha'(G) \quad \text{for every graph } G \text{ of order } n \text{ without } C_4,$$

$\alpha'(G)$ being the matching number. Immediately after [1, Corollary 36] they ask:

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Open problem 37 of Bousquet, Deschamps, Lehtilä and Parreau.

Does there exist a graph G without C_4 as a subgraph with $\text{mold}(G) < n - \alpha'(G)$?

The statement is quoted from the e-print [arXiv:2112.01910v2](#) of [1], where it is numbered Open problem 37. In [1] “without H ” and “ H -free” are different, the latter being reserved for the induced version, so the hypothesis of the problem is that G contains no 4-cycle at all.

Theorem 1. *Let G be the graph on $V = \{0, 1, \dots, 12\}$ with graph6 string `L??E@agXAp@wdC`, that is, with the 22 edges*

$$\begin{aligned} &\{0, 6\}, \{0, 8\}, \{0, 12\}, \{1, 6\}, \{1, 9\}, \{1, 10\}, \\ &\{2, 7\}, \{2, 8\}, \{2, 9\}, \{3, 7\}, \{3, 10\}, \{3, 12\}, \\ &\{4, 8\}, \{4, 10\}, \{4, 11\}, \{5, 9\}, \{5, 11\}, \{5, 12\}, \\ &\{6, 11\}, \{7, 11\}, \{8, 10\}, \{9, 12\}. \end{aligned}$$

Then G has no C_4 subgraph, $n = 13$, $\alpha'(G) = 6$ and $\text{mold}(G) = 6$. In particular

$$\text{mold}(G) = 6 < 7 = n - \alpha'(G),$$

so Open problem 37 of [1] has an affirmative answer.

2. THE GRAPH, AND ITS MATCHING NUMBER

The adjacency lists of G are

$$\begin{array}{llll} 0 : 6, 8, 12 & 1 : 6, 9, 10 & 2 : 7, 8, 9 & 3 : 7, 10, 12 \\ 4 : 8, 10, 11 & 5 : 9, 11, 12 & 6 : 0, 1, 11 & 7 : 2, 3, 11 \\ 8 : 0, 2, 4, 10 & 9 : 1, 2, 5, 12 & 10 : 1, 3, 4, 8 & 11 : 4, 5, 6, 7 \\ 12 : 0, 3, 5, 9 & & & \end{array}$$

so G is connected with eight vertices of degree 3, five of degree 4, and $\sum_v \deg v = 44 = 2 \cdot 22$. The edge list above is written in lexicographic order and we keep that order throughout: an *orientation index* is an integer o with $0 \leq o < 2^{22}$, and bit i of o is 1 exactly when the i -th listed edge $\{a, b\}$ (with $a < b$) is oriented $a \rightarrow b$. Thus bit 0 governs the edge $\{0, 6\}$ and bit 21 the edge $\{9, 12\}$.

Lemma 2. *G has no C_4 subgraph.*

Proof. A 4-cycle $uxvy$ consists of two distinct vertices u, v together with two distinct common neighbours x, y of them, and conversely any two distinct vertices with two distinct common neighbours span a 4-cycle. The pair u, v need *not* be adjacent, so the condition to check is that every one of the $\binom{13}{2} = 78$ unordered pairs of vertices has at most one common neighbour, not merely each of the 22 adjacent ones. Inspection of the thirteen adjacency lists above shows that it holds: 54 of the 78 pairs have exactly one common neighbour and the other 24 have none, in agreement with the count $\sum_v \binom{\deg v}{2} = 8 \binom{3}{2} + 5 \binom{4}{2} = 54$ of paths of length two. $\square$

Lemma 3. $\alpha'(G) = 6$, hence $n - \alpha'(G) = 7$; and $A = \{0, 1, 2, 3, 4, 5\}$ is independent in G .

Proof. The six edges $\{0, 6\}, \{1, 9\}, \{2, 7\}, \{3, 10\}, \{4, 11\}, \{5, 12\}$ are listed above and have twelve distinct endpoints, so they form a matching and $\alpha'(G) \geq 6$. Since $n = 13$ is odd, $\alpha'(G) \leq \lfloor 13/2 \rfloor = 6$. Hence $\alpha'(G) = 6$ and $n - \alpha'(G) = 13 - 6 = 7$. Finally, none of the fifteen pairs inside A appears in the edge list, so A is independent. $\square$

3. $\text{mold}(G) \geq 6$, FROM A SINGLE ORIENTATION

Proposition 4. $\text{mold}(G) \geq 6$.

Proof. Let $A = \{0, 1, 2, 3, 4, 5\}$, independent by Lemma 3. Exactly 18 of the 22 edges have an endpoint in A – all of them but $\{6, 11\}, \{7, 11\}, \{8, 10\}$ and $\{9, 12\}$ – and each of those 18 has its other endpoint in $B = \{6, \dots, 12\}$. Let D^* be any orientation of G in which all 18 of those edges are directed *away* from A ; the remaining four edges, which lie inside B , may be oriented arbitrarily, so there are 16 such orientations and the argument below applies to each.

In D^* every vertex of A is a source, i.e. $N_{D^*}^-(u) = \emptyset$ for $u \in A$. Hence if S is locating-dominating in D^* and $u \in A$ were outside S , then $I_{D^*}(u) = \emptyset$, which is forbidden. So $A \subseteq S$ and $\gamma_{\text{LD}}(D^*) \geq |A| = 6$.

Conversely $S = A$ is locating-dominating in D^* : the seven traces are

$$I(6) = \{0, 1\}, \quad I(7) = \{2, 3\}, \quad I(8) = \{0, 2, 4\}, \quad I(9) = \{1, 2, 5\},$$

$$I(10) = \{1, 3, 4\}, \quad I(11) = \{4, 5\}, \quad I(12) = \{0, 3, 5\},$$

that is, $I(v) = N(v) \cap A$ for each $v \in B$. They are non-empty, and they are pairwise distinct: three of them have size 2 and four have size 3, and the three of size 2 are pairwise distinct, as are the four of size 3. Hence $\gamma_{\text{LD}}(D^*) = 6$ and $\text{mold}(G) \geq 6$. $\square$

4. $\text{mold}(G) \leq 6$: THE ONE FINITE COMPUTATION

Lemma 5 (monotonicity). *Let D be an orientation of a graph and let $S \subseteq S' \subseteq V(D)$. If S is locating-dominating in D , so is S' .*

Proof. Let $v \in V(D) \setminus S'$. Then $v \notin S$ and $I'(v) := N_D^-(v) \cap S' \supseteq N_D^-(v) \cap S = I(v)$, which is non-empty, so $I'(v) \neq \emptyset$. Moreover $S \subseteq S'$ gives $I'(v) \cap S = I(v)$ for every such v ; hence if $I'(u) = I'(v)$ for two distinct $u, v \notin S'$ then $I(u) = I'(u) \cap S = I'(v) \cap S = I(v)$, contradicting that S is locating-dominating. No hypothesis on the graph is used. $\square$

In particular, for $k \leq n$ the inequality $\gamma_{\text{LD}}(D) \leq k$ holds if and only if some set of size *exactly* k is locating-dominating in D , so it is enough to examine sets of one size.

Proposition 6. *Every one of the $2^{22} = 4\,194\,304$ orientations of G admits a locating-dominating set of size 6. Hence $\text{mold}(G) \leq 6$, and with Proposition 4, $\text{mold}(G) = 6$.*

This is a finite statement, but not hand-sized: the number of orientations times the $\binom{13}{6} = 1716$ candidate six-sets is about 7×10^9 pairs. Here is the reduction used.

Fix a 6-set S and put $T = V \setminus S$, $|T| = 7$. Whether S is locating-dominating in an orientation D depends on D only through the seven traces $I_D(v) = N_D^-(v) \cap S$, $v \in T$; and $I_D(v)$ is determined by the directions of the edges joining v to S . Those edge sets, over $v \in T$, are pairwise disjoint, and edges inside S or inside T are irrelevant. Inside the 2^{22} -element space of orientation indices:

- (a) “ $I_D(v) = \emptyset$ ” is the set of indices agreeing with one prescribed pattern on the $|N(v) \cap S|$ bits of the edges from v to S , and free on the other $22 - |N(v) \cap S|$ bits;
- (b) “ $I_D(u) = I_D(v)$ ” for $u \neq v$ in T is the union, over the non-empty subsets $c \subseteq N(u) \cap N(v) \cap S$, of the sets of indices agreeing with one prescribed pattern on the bits of the edges from u to S and from v to S . By Lemma 2 the set $N(u) \cap N(v)$ has at most one element, so there is at most one such c and the union has at most one term.

Each of the sets in (a) and (b) is a subcube of the orientation cube: the indices agreeing with a prescribed pattern on a set F of bit positions are obtained from the single index carrying that pattern by the $22 - |F|$ doublings $X \mapsto X \cup (X + 2^q)$, one per free position q , each a shift-and-or on a 2^{22} -bit word. So for fixed S the set of orientations in which S is *not* locating-dominating is a union of at most $7 + \binom{7}{2} = 28$ such subcubes; unioning the complements over the six-sets S and comparing with the all-ones mask decides Proposition 6 exactly.

Carrying this out, 980 of the 1716 six-sets already suffice: the union of their masks is all of $\{0, 1, \dots, 2^{22} - 1\}$. Proposition 6 follows, and hence Theorem 1.

5. SCOPE

Existence is all that is claimed. We do not claim that G is of least order or least size, nor that it is unique, and nothing here shows that 13 is the least order at which the answer is affirmative. The bound $\text{mold}(G) \leq n - \alpha'(G)$ of [1, Lemma 34] remains a theorem; we show only that it is not always tight, and we do not determine for which graphs without C_4 it is tight.

Neighbouring settings: locating-dominating sets in graphs of girth at least 5 [4], a subclass of the graphs without C_4 that does not contain G , since G has the triangle $\{4, 8, 10\}$; location-domination together with matchings in undirected cubic graphs [6]; the *minimum* of the identifying-code number over all orientations [5]; best and worst orientations for the metric dimension [2]; and γ_{LD} of local tournaments [3].

Lemmas 2 and 3 and Proposition 4 need no machine. A program that carries out the computation of Proposition 6 from the graph6 string alone by the reduction of Section 4, together with a recorded run of it, accompanies this note; the program also re-derives the statements proved above by hand and cross-checks the mask construction against the naive per-orientation predicate.

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